

YIFEI MA

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Shanghai, China

YIFEI-MA

PROFESSIONAL EXPERIENCE

Software Engineer

KargoBot (DiDi Voyager Autonomous Trucking)

Sep 2021 – Present

Shanghai, China

Middleware Development for Automated Vehicles

- Co-architected high-availability middleware with DDS-compatible communication layer supporting hybrid networking and data replay
- Optimized message scheduling algorithms and memory management, reducing average end-to-end latency by $\geq 20\text{ms}$ and P9999 latency by $\geq 150\text{ms}$
- Led migration of 10+ core nodes to proprietary middleware, enhancing overall system performance and reliability
- Developed and optimized core components including message queues and timing wheels, leveraging lock-free queues and shared memory technologies to support high-performance, high-concurrency, and real-time requirements

Distributed Performance Monitoring Platform

- Built distributed and comprehensive online/offline performance analytics for L4 vehicles with 30+ core metrics visualization
- Integrated Perfetto + eBPF + ftrace for full-stack tracing from application to kernel, improving bottleneck identification efficiency by 3x
- Established multi-version performance baseline comparison and automated alerting, integrating road test and simulation data into CI/CD for high-quality releases

Sensor Driver Architecture Redesign & Optimization

- Implemented unified drivers supporting multiple LiDAR/Radar models with real-time health monitoring, alerting, and auto-recovery mechanisms
- Achieved 80% CPU usage reduction through lock-free queues, zero-copy, and SIMD optimizations; reduced data latency by 35ms via optimized frame slicing strategy

High-Performance Data Compression & Storage System

- Developed ZSTD streaming compression-based PCAP collection system, compressing LiDAR data to 17% of ROS bag size, significantly reducing I/O overhead
- Optimized image storage using HEVC encoding, achieving 40% size reduction in All-intra mode
- Built compressed-format data replay simulation system enabling data closed-loop, saving \$9M annually in storage costs and supporting fleet scaling from 50 to 300+ vehicles

Intelligent Simulation & Testing System

- Developed ViL/HiL hybrid simulation platform with 20+ automated evaluation metrics and local simulation visualization toolchain with fault injection capabilities
- Built simulation consistency validation tools and deterministic simulation framework ensuring algorithm iteration quality

TECHNICAL SKILLS

ROS/DDS gRPC/Protobuf C++ Python eBPF/ftrace
Docker CUDA/Parallel Computing Distributed Systems
Qt PyTorch OpenCV

EDUCATION

B.Sc. Computing Science (Honors)

University of Alberta

Sep 2017 – Jun 2021 Canada

- GPA: Top 10% (Dean's Honor Roll)
- Core Courses: Operating Systems, Computer Vision, Machine Learning, Natural Language Processing, Distributed Systems
- Projects: Video Moving Object Segmentation, MCTS-based Go AI, Android Ride-sharing App

LANGUAGES

Chinese

English



INTERESTS

Photography

Go (3-dan)

Basketball

KEY PROJECTS

Low-Latency Video Streaming System

Core Feature on E-Stop System

Jun 2024 – Sep 2024

- Developed accelerated multi-channel 4K video processing pipeline (undistortion $\rightarrow$ stitching $\rightarrow$ RTSP streaming)
- Achieved low overhead (single CPU core $< 40\%$) through CUDA zero-copy + AV1/HEVC hardware encoding
- Delivered real-time video transmission with GTG latency $\leq 200\text{ms}$ for low-bandwidth radio communication

Autonomous Driving Simulation Smart Agent

Tsinghua University

Sep 2022 – Jun 2023

- Collaborated on intelligent traffic participant system development for Virtual Sim platform
- Implemented hybrid multi-directional avoidance and path planning algorithms based on rule-based, social force models, and reinforcement learning
- Enhanced simulation scenario realism and interactivity, supporting large-scale algorithm validation and testing